

Supplementary Material for: Pitch, Prediction, and Speaker Confounds in U.S. Presidential Debates, 1988-2020

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Abstract

This document is the full supplementary material for the paper above: the debate inventory, alignment and preprocessing detail, complete feature definitions, full statistical results, the discourse-annotation audit, supplementary multimodal models, LLM methods and reproducibility specification, the exploratory same-candidate analysis, and known corpus issues. The main paper’s appendix gives a short summary of each section and points here for the full content. Section and table numbers here are independent of the main paper’s numbering (prefixed S). Code and data are available at: <https://github.com/researchsparkhub/us-presidential-debate-pitch-confounds>

S1 Debate Inventory

Table S1: Every debate in the corpus: date, candidates, and which ticket won the election that followed.

ID	Date	Candidates	Winner
1001	1988-09-25	G. H. W. Bush vs. Dukakis	Bush
1003	1988-10-13	G. H. W. Bush vs. Dukakis	Bush
1004	1992-10-11	Clinton vs. Perot, G. H. W. Bush	Clinton
1006	1992-10-15	Clinton vs. Perot, G. H. W. Bush	Clinton
1007	1992-10-19	Clinton vs. G. H. W. Bush, Perot	Clinton
1008	1996-10-06	Clinton vs. Dole	Clinton

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Table S1 continued

ID	Date	Candidates	Winner
1010	1996-10-16	Clinton vs. Dole	Clinton
1011	2000-10-03	G. W. Bush vs. Gore	Bush
1013	2000-10-11	G. W. Bush vs. Gore	Bush
1014	2000-10-17	G. W. Bush vs. Gore	Bush
1015	2004-09-30	G. W. Bush vs. Kerry	Bush
1017	2004-10-08	G. W. Bush vs. Kerry	Bush
1018	2004-10-13	G. W. Bush vs. Kerry	Bush
1019	2008-09-26	Obama vs. McCain	Obama
1021	2008-10-07	Obama vs. McCain	Obama
1022	2008-10-15	Obama vs. McCain	Obama
1023	2012-10-03	Obama vs. Romney	Obama
1025	2012-10-16	Obama vs. Romney	Obama
1026	2012-10-22	Obama vs. Romney	Obama
1027	2016-09-26	Trump vs. H. Clinton	Trump
1029	2016-10-09	Trump vs. H. Clinton	Trump
1030	2016-10-19	Trump vs. H. Clinton	Trump
1031	2020-09-29	Biden vs. Trump	Biden
1033	2020-10-22	Biden vs. Trump	Biden

S2 Alignment and Preprocessing

Audio was converted to single-channel PCM at 16 kHz. For matching, both the recognised text and the official transcript were reduced to lower-case alphanumeric tokens. Punctuation was removed, contractions were left unexpanded, and numerals were kept as written. The match rate is the share of official-transcript tokens that fall inside a matching block. Matching blocks were found with a longest-matching-block algorithm. Routing thresholds were $\geq 85\%$ and 60-85%.

Recognition used Whisper (Radford et al., 2023), which gives word-level timings. Matched tokens take those timings directly. Unmatched runs are interpolated between the nearest matched words, split by character count.

Montreal Forced Aligner (McAuliffe, Socolof, Mihuc, Wagner, & Sonderegger, 2017) was run with the English ARPA acoustic model and dictionary. Default beam settings did not finish on ninety-minute recordings, so production runs used wider beams under a watchdog. It converged on 20 of 24 debates and is kept as an independent check.

S3 Complete Feature Definitions

Pitch, loudness and voice-quality measures come from Praat (Boersma & Weenink, 2024) through `parselmouth`. The 88-parameter set comes from openSMILE (Eyben et al., 2016; Eyben, Wöllmer, & Schuller, 2010), which also supplies the semitone pitch measures and the second voice-clarity implementation. Parsing and tagging use spaCy (Honnibal, Montani, Van Landeghem, & Boyd, 2020) with `en_core_web_sm`, the smallest available model. Its weakest outputs are parse depth and dependency length. Sentiment uses VADER (Hutto & Gilbert, 2014), topic categories use Empath (Fast, Chen, & Bernstein, 2016), and sentence vectors use Sentence-BERT (Reimers & Gurevych, 2019).

A silent pause is a stretch of at least 150 ms during which loudness stays at least 25 dB below the ninetieth percentile for that sentence. The usual threshold after [Goldman-Eisler \(1968\)](#) is 250 ms. The detector cannot see filled pauses. Whether those were transcribed at all varies across this corpus.

Table S2 names the 34 standardized measures used in the combined multimodal models and the LLM protocol. These overlap heavily with the full 43-measure broad screen. The remaining measures are named individually in the main paper’s results: jitter, shimmer, four readability scores, positive/negative sentiment, and the 35 topic categories, which are treated as one diagnostic family rather than as individual screened measures.

Table S2 The 34 standardized measures used in the combined and LLM analyses, by family.

Acoustic (Praat)	Acoustic (openSMILE)	Linguistic
Mean pitch	Syllable rate	Clauses
Pitch SD (Hz)	Articulation rate (syll/s)	Subordinate clauses
Pitch range (Hz)	Semitone pitch SD/mean	Parse depth
Loudness	Semitone 20-80 range	Dependency length
Voice clarity (Praat)	Voice clarity (openSMILE)	Words per sentence
Pause duration		Word length
Pause count		Function words
Mean pause length		1st/2nd/3rd-person pronouns
Speech rate		Nouns, verbs, adjectives, adverbs
Articulation rate (w/s)		Reading ease, grade level
		Sentiment, positive, negative

For the frame-level pitch-contour measures, the low-level-descriptor output of openSMILE (F0semitoneFrom27.5Hz_sma3nz, 10 ms hop) was run independently per sentence clip. We first tried slicing these measures from a continuously smoothed whole-debate stream, but rejected this approach: continuous smoothing across sentence boundaries shifted the resulting statistics measurably at clip edges. Distributional measures use all voiced frames regardless of order. Dynamic measures use only differences between temporally consecutive voiced frames. Declination slope and its residual standard deviation come from an ordinary-least-squares fit of the semitone contour against frame time.

Four measurement problems are recorded for anyone reusing the corpus. *One pitch range for everyone*: the analysis range was 75-500 Hz for a corpus where average pitch runs from 102.5 to 195.9 Hz. The fixed 75 Hz floor is the likely cause of the pitch-dependence of voice clarity. *Mismatched floors*: Praat’s loudness default assumes a minimum pitch of 100 Hz against the 75 Hz used for pitch. *Voiced fraction* runs from 0.583 to 0.742 across speakers. It correlates with voice clarity at +0.674 at speaker level, and this relationship persists within a single speaker in a single debate, at a median of +0.327. *Reliability depends steeply on sentence length* (Table S3).

Table S3 Voice-clarity reliability by sentence length. Split-half reliability is odd-versus-even sentences within one speaker in one debate.

Sentence length	Mean clarity	SD	Split-half
0.3-1.0 s	8.79 dB	3.10	0.126
1.0-2.5 s	9.51 dB	2.42	0.476
2.5-6.0 s	9.96 dB	2.16	0.655
6.0 s+	10.30 dB	2.00	0.729
Candidate sentences: median 3.02 s, IQR (1.53, 5.71); 24.2% under 1.5 s, 13.1% under 1.0 s, 4.6% under 0.5 s.			

S4 Full Statistical Results

Debate effects are removed by subtracting each debate’s mean. Cluster-robust variances use a sandwich estimator with a small-sample correction. These are referred to a t distribution with $G - 1$ degrees of freedom: $G = 24$ for debate clustering and 14 for speaker clustering. With this few clusters the p -values should be read as generous rather than exact.

The randomization test enumerates all $2^{24} = 16,777,216$ ways of flipping the winner label within debates. Random-effects pooling follows DerSimonian-Laird. The prediction interval is computed as the pooled estimate $\pm t_{22}\sqrt{\tau^2 + \text{SE}^2}$. Leave-one-speaker-out recomputes the within-debate effect with one speaker’s sentences removed.

Several of the 43 measures overlap: 13 pairs correlate above $|r| = 0.7$. The effective number of independent measures is 30.0 by the Li-Ji method. Correcting at $m = 30$ instead of $m = 43$ changes nothing.

The area under the curve (AUC) reported for the discriminant and forest in the main paper has a direct reading: take one sentence from a winning candidate and one from a non-winning candidate at random, and AUC is the probability the model scores the winning one higher. 0.5 is the right baseline only when the held-out items are independent.

S5 Discourse Annotation

The rule-based labeller assigns dialogue acts and rhetorical categories from sentence type, word and pattern matches, question syntax, and first-person stance markers. Two of its categories are copies of the sentence-type column rather than independent detectors. It returns zero for 13 codes. Table S4 shows what fraction of each category’s hits come from a single string.

Fixing the four fixable rules—anchoring at word boundaries and requiring a real unity or patriotic phrase—cuts unity framing from 396 hits to 69. It also cuts patriotic appeal from 1,235 to 115. Between 94 and 97 per cent of what these categories counted was something else. What survives appears in 0.3 to 0.5% of sentences with differences no larger than 0.014.

Table S4 What the rule-based detectors actually fire on. “Sole trigger” is the share of hits produced by one string and nothing else. “Rep.” is the count after fixing the rule. “Eff.” is the difference after fixing.

Category	Hits	Sole trigger	Rep.	Eff.	dir.
Unity framing	396	“United States” 80.1%	69	−0.004	10/24
Patriotic appeal	1,235	“America” 88.0%	115	−0.014	13/24
Historical ref.	265	bare year 63.4%	165	−0.028	15/24
Fear appeal	540	substring 30.9%	373	+0.007	11/24
Emotional appeal	435	“children” + “families”	–	–	–
Agreement	237	“absolutely” / “exactly”	–	–	–
Rejection	178	initial “no” 93.8%	–	–	–
Statement	22,629	= declarative 100%	–	–	–
Imperative	697	= imperative form 100%	–	–	–

The corpus was also labelled by a large language model through a batch API, using a prompt that assigns dialogue acts first and rhetorical categories second. The model returned a parseable answer for 100.0% of sentences. It assigned at least one dialogue act to 99.86% of candidate sentences and at least one rhetorical category to 71.05%. Table S5 gives category-level agreement. The two labellings agree best on categories readable from the grammar (questions, thanks), and worst on the interpretive ones.

Table S5 Agreement between the rule-based and model labels over 24,193 candidate sentences. PSA (percent specific agreement) treats neither as the reference. PABAK and AC1 correct for how rare each category is.

Category	Rule	Model	PSA	κ	PABAK	AC1
Thanks	182	217	0.852	0.851	0.995	0.998
Wh-question	212	376	0.599	0.594	0.980	0.990
Polar question	146	290	0.564	0.561	0.984	0.992
Explanation	1,478	869	0.415	0.387	0.886	0.937
Personal stance	1,424	4,865	0.389	0.327	0.682	0.795
Fear appeal	540	1,674	0.237	0.210	0.860	0.923
Historical ref.	265	1,343	0.208	0.193	0.895	0.944
Patriotic appeal	1,235	282	0.190	0.174	0.898	0.946
Imperative	697	1,234	0.177	0.146	0.869	0.929
Unity framing	396	1,099	0.060	0.037	0.884	0.938
Rebuttal	3	4,653	0.001	0.001	0.616	0.767
Informative stmt.	2,475	0	0.000	0.000	0.795	0.887
Whole-set agreement (Krippendorff α , MASI): dialogue acts +0.199; rhetorical categories +0.012.						

Chain-of-thought variant. A natural objection to the direct prompt above is that it never gives the model room to reason. It goes straight from sentence to codes. We re-ran the labelling on a stratified sample of 3,001 candidate sentences (12.4% of the corpus, proportional across all 24 debates), with one change: the output schema requires a short free-text **reasoning** field, written before the **dams1** and **beads** codes. This means the model must commit to an explanation before it commits to a label. It is the same chain-of-thought design that Wei et al. (2022) showed can improve

accuracy on tasks with a checkable right answer. The model, codebook, and prompt were otherwise the same (Section S7). It does not help. Figure S1 gives Krippendorff’s α against the rule tagger for both prompting styles, recomputed on the identical sample. Dialogue acts move from +0.167 direct to +0.169 with reasoning. Rhetorical categories move from -0.030 to -0.002 . Both changes are within sampling noise of each other, and both are the same chance-level result as the full-corpus comparison above. Chain-of-thought and direct prompting agree with *each other* at $\alpha = +0.564$ for dialogue acts and $+0.536$ for rhetorical categories. This is moderate agreement, well short of the near-unanimity two runs of a deterministic rule tagger would show. Both prompting styles remain equally uncorrelated with the rule-based labels. Reasoning also makes the model slightly more conservative: it assigns a dialogue act to 98.20% of the sample and a rhetorical category to 66.61%, against 99.83% and 70.68% for direct prompting on the identical sample. The instrument’s problem was never that the model failed to explain itself. The problem is that the categories it is asked to detect are not observable from either the grammar or an explanation. Asking for one does not manufacture the signal.

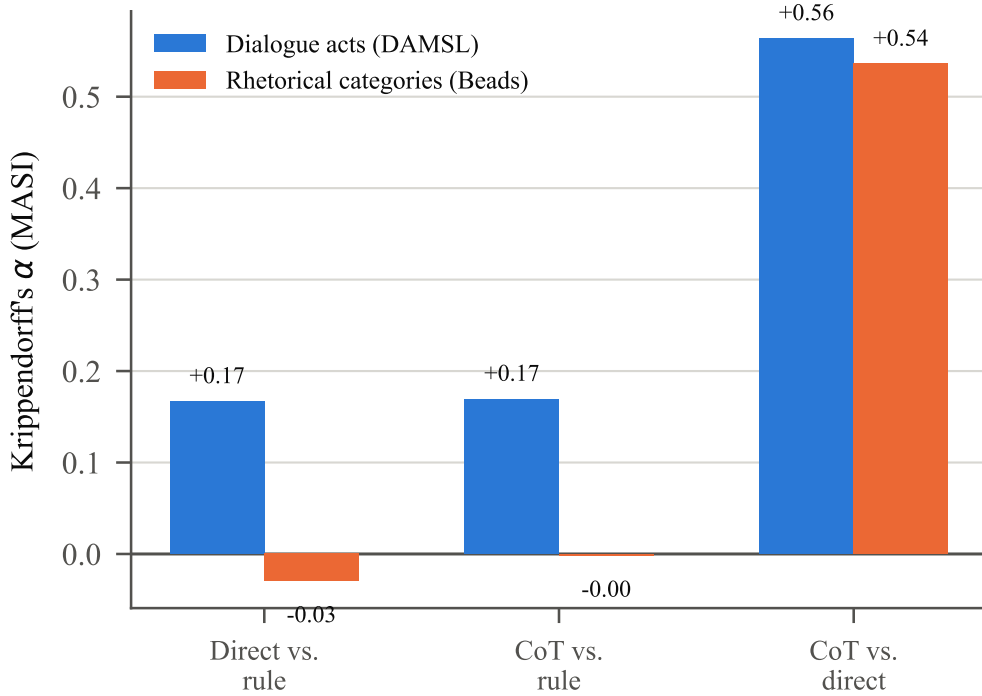


Fig. S1 Chain-of-thought prompting does not close the gap with the rule-based tagger. Both prompting styles land at the same chance-level agreement with the rules; what changes is how much the two LLM conditions agree with each other.

S6 Supplementary Multimodal Models

Weighted sum. Each measure gets a weight equal to how well it separates the groups in training debates T :

$$w_j = \frac{1}{|W_T|} \sum_{i \in W_T} z_{ij} - \frac{1}{|L_T|} \sum_{i \in L_T} z_{ij}, \quad (\text{S1})$$

The weight vector is scaled to unit length. Each sentence is then scored as $S_i = \sum_{j=1}^{34} w_j z_{ij}$. Weights are learned on 23 debates and scored on the twenty-fourth, rotating through all of them.

Table S6 Combining measures, against the two measures that survived on their own.

	Single measures		Composite (34 measures)	
	Semitone 20-80 range	Semitone SD/mean	hold out debate	hold out speakers
Mean per-debate effect	-0.243	-0.370	+0.390	+0.070
SD across debates	0.289	0.358	0.603	0.519
Debates in direction	21/24	20/24	17/24	11/24
Sign-test p	0.0003	0.0015	0.064	0.84

On held-out debates, the weighted composite reaches +0.390 standard deviations, a 60% improvement on the semitone percentile range alone. But it is markedly less consistent (SD across debates 0.603 against 0.289). Under speaker holdout it falls to +0.070. A single measure needing no training is more reliable than a weighted combination of 34.

Discriminant loadings. Averaged to the 51 speaker-by-debate cells, the discriminant separates winning from non-winning cells at an area under the curve of 0.907. Table S7 reports the correlation between each measure and the discriminant score. Correlations are more stable than raw weights when measures are correlated.

Every measure in the top ten is acoustic. Nothing syntactic, lexical or semantic appears until well below. The highest-loading measure is the one shown in the main paper to track the speaker’s own pitch at $r = +0.78$ to $+0.90$. Consistent with that, 81.0% of the variance in the cell scores is explained by which speaker the cell belongs to.

Principal component analysis. PCA asks whether the outcome happens to line up with the largest-variance directions among the 34 measures without using the labels at all.

PC1 (20.9% of variance) is a sentence-length dimension separating the groups at +0.006 standard deviations—nothing. PC2 is speaking rate. The component that best separates the groups is PC5, at +0.359, accounting for only 6.4% of the variance. A

Table S7 What the discriminant leans on. Positive means higher among winning candidates.

Measure	Loading
Voice clarity (openSMILE)	+0.628
Semitone pitch SD/mean	−0.587
Voice clarity (Praat)	+0.462
Pitch SD (Hz)	−0.422
Semitone 20-80 pitch range	−0.410
Pitch range (Hz)	−0.317
Articulation rate (syll/s)	+0.235
Mean pitch	+0.189
Pause duration	+0.186
Articulation rate (words/s)	+0.181

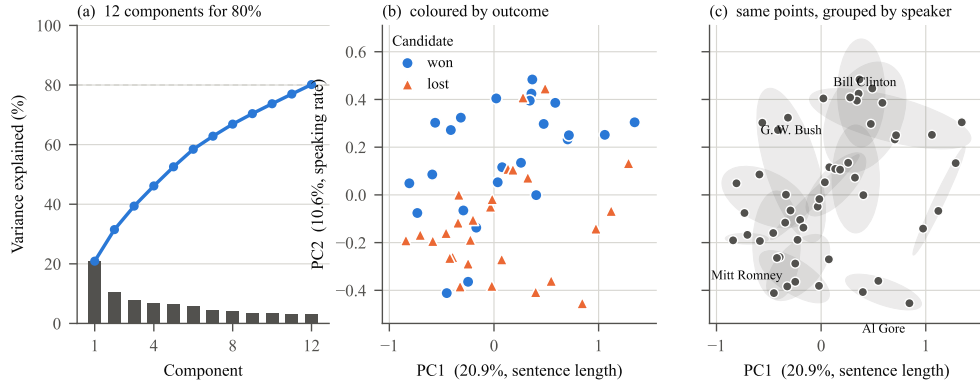


Fig. S2 Principal component analysis of the 34 standardized measures. (a) Twelve components are needed for 80% of the variance. (b) Each point is one candidate in one debate, coloured by outcome. (c) The same points, with an ellipse around each speaker’s debates. The apparent separation by outcome in (b) coincides with the speaker grouping in (c).

Table S8 Principal components of the 34 standardized measures.

	% of variance	Separation of groups	Dominant measures
PC1	20.9	+0.006	Sentence length, grade level
PC2	10.6	+0.136	Articulation rate, speech rate
PC3	7.9	+0.203	Pitch SD, reading ease
PC4	6.8	+0.070	Mean pitch, voice clarity
PC5	6.4	+0.359	Voice clarity (both tools)
PC6	5.9	+0.044	Sentiment

12 components needed for 80% of variance.
At cell level, speaker explains 83.6% of PC1 and 86.6% of PC2 positions.

hand-built weighted sum, a supervised discriminant, and an unsupervised decomposition that ignores the labels entirely all reach the same place: what structure exists in these 34 measures is mostly about who is speaking.

LLM “both” condition. Given the transcript and the descriptor table together, the model calls 24 of 24 debates correctly. This agrees with the transcript-only condition (main paper) on all 24 debates. Adding numeric descriptors to a transcript the model can already place in history changes nothing, because the transcript had already settled the answer.

S7 LLM Methods and Reproducibility

This appendix specifies every language-model call in the paper: the original discourse tagging and its chain-of-thought variant (Section S5), and the cross-modal winner prediction with its contamination probe. The goal is that another researcher could reissue every call from this description alone.

Model. All components use the same model, `claude-opus-4-8`, accessed through Anthropic’s Messages API. Anthropic does not publish a parameter count for this model, so none is reported here. Reporting one would imply a precision we do not have. Every call uses structured (`json_schema`) output: the model’s response is constrained at generation time to match a schema we specify, rather than parsed after the fact from free text. All calls were issued on 2026-08-06.

Sampling. This model deprecates the `temperature` parameter. The API rejects a request that sets it. Repeated calls therefore draw on whatever sampling variability the API applies by default, not on a caller-chosen temperature. Where the paper reports a repeat count, that is the only lever available for measuring stability.

Discourse tagging. System prompt, codebook and full JSON schema are reproduced verbatim in `Sahana Project/tagging_prompt.md` (direct) and `Sahana Project/cot_tagging_prompt.md` (chain-of-thought) in the code repository. Both send 10 sentences per request, batched for throughput, with the system prompt and codebook cached (`cache_control: ephemeral`) across requests. The only schema difference is one field, `reasoning`, which the chain-of-thought variant requires before `dams1` and `beads`. Structured output fills schema fields in the order they are declared. This forces the free-text reasoning to be generated before the codes rather than after them.

Cross-modal winner prediction. System prompt, both JSON schemas (descriptor-citing and theme-citing), the contamination-probe prompt and schema, and the closed evidence vocabularies are reproduced verbatim in `Sahana Project/crossmodal_prompt.md` in the code repository. One call is issued per debate, per condition, per repeat. Each repeat uses a freshly randomised Candidate-A/B/C letter assignment, shared across conditions within that repeat so predictions are comparable trial-for-trial. The contamination probe issues one further call per debate per input type, with no repeats. It is a single-shot identification question rather than a graded judgement. Every call uses a fresh, empty context, so agreement or disagreement between conditions reflects the input alone.

Anonymisation. Candidate names, party affiliation and the debate date are removed from every input. Moderator turns are dropped entirely, leaving only candidate sentences in original chronological order. This is a partial defence at best. A transcript’s content-quoted lines, named policies, the other candidate’s real name

spoken aloud mid-turn—was not scrubbed. Removing it would have taken out the communicative content the transcript condition exists to test. The contamination probe measures what leaks through this partial anonymisation rather than assuming it is complete. All 24 of 24 transcripts were identified regardless of the redaction.

Grounding check. A cited measure counts as consistent with a debate’s real data when the candidate it names as favoured actually has the higher within-debate standardized value for that measure. A margin of less than 0.15 standard deviations between the two candidates is scored as “similar” and excluded rather than forced to a side. This is an automatic, closed-vocabulary check, not a human judgement of whether the reasoning sounds plausible.

S8 Exploratory Same-Candidate Analysis

Table S9 Where the same speaker lands when they won and when they lost. The discriminant was fitted *without* that speaker, then their debates were projected in. Scores are relative to the opponent on the night.

Speaker	Debate	Outcome	Score
G. H. W. Bush	1988, first	won	−0.021
	1988, second	won	−0.066
	1992, first	lost	−0.224
	1992, second	lost	−0.203
	1992, third	lost	−0.223
	Gap (won − lost)		+0.173
	Winning above losing		6/6
Donald Trump	2016, first	won	+0.010
	2016, second	won	−0.018
	2016, third	won	−0.025
	2020, first	lost	−0.034
	2020, second	lost	+0.002
	Gap (won − lost)		+0.005
	Winning above losing		4/6

S9 2024 Exclusion and Known Corpus Issues

Two 2024 debates exist in the corpus but are held out, because we could not verify who was speaking. Their official transcripts are paraphrased news summaries, sharing only 5.2% and 5.5% of their words with the audio. As a result, the recognised text was used as the transcript, and speakers were assigned by diarization (Bredin, 2023). That diarization was never checked against hand-labelled data. We have no error rate for it.

Four things prompted the exclusion. In the June 2024 debate one candidate is credited with 786 sentences and the other with 256—three to one, in a format with timed turns and muted microphones. In the September debate about 285 sentences

sit in twelve or more leftover speaker clusters. Across the pair, 1,162 sentences are labelled *audience*, in two debates with no audience. Finally, candidates account for 43.9% of sentences in these two, against 76 to 86% everywhere else.

The confirmation is what happens on removal: across the remaining 24 debates, unattributed sentences drop to zero and audience sentences to 287. All of the unattributed material and 80.2% of the audience material came from those two debates. Including the pair changes several results. Noun rate flips sign, from -0.012 to $+0.043$. Sentiment roughly halves. The embedding axis rotates (cosine similarity 0.895 to the one reported here). Finally, 19 of the 50 most extreme sentences at one end come from this 9.5% of the data. These transcripts also contain no truncation marks and no filled pauses at all. As a result, interruption and self-correction are missing by construction.

Availability

The corpus is deposited as code and derived data at <https://github.com/researchsparkhub/us-presidential-debate-pitch-confounds>. Released material covers the annotation layers, the per-sentence features and the checking scripts. Alignment scripts point at audio sourced elsewhere, which cannot be redistributed; see that repository’s `DATA_AVAILABILITY.md` for how to source it.

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